

Stacked Ensemble Learning for Residential Appliance Energy Prediction and Anomaly Screening

Rahul Gupta, Umang Gupta, Garv Gulati, Nikhil Goyal

Department of Computer Science and Engineering, SRM Institute of Science and Technology, Delhi NCR Campus
Modinagar, Ghaziabad, India

EMAIL ID:

Abstract

Residential appliance energy varies with household routines and environmental conditions, complicating prediction at short intervals. This study examines a stacked ensemble for nominal-demand prediction and complementary anomaly screening using the public Appliances Energy Prediction dataset. The source contains 19,735 ten-minute observations from one Belgian dwelling. Temporal and environmental descriptors feed Extra Trees, XGBoost, and histogram gradient boosting regressors, whose predictions are combined through a non-negative ridge layer. Isolation Forest and residual thresholds provide distinct screening signals. The documented experiment retains 16,740 observations with appliance energy between 10 and 120 Wh and states a test sample of 5,022 observations. Reported preliminary performance is $R^2 = 0.7504$, mean absolute error = 8.62 Wh, root mean squared error = 11.58 Wh, and mean absolute percentage error = 15.53%. The diagnostic figures identify 24 flagged observations, equivalent to 0.48% of the stated test sample. These flags indicate unusual prediction errors rather than verified appliance faults. The evaluation split, complete feature schema, stacking conventions, and screening calibration remain incompletely specified. Accordingly, the results support a preliminary assessment of an integrated architecture, without establishing superiority over matched baselines or generalization to unrestricted household demand. A proposed REST interface connects prediction and retrospective auditing. Future validation requires chronological evaluation, independent threshold calibration, reproducible prediction records, and testing across households.

Keywords: Residential appliance energy; Stacked ensemble learning; ExtraTrees; Non-negative ridge regression; Anomaly screening

1. Introduction

Residential appliance demand changes abruptly as cooking, washing, lighting, and other activities overlap. Similar environmental conditions can therefore coincide with different consumption levels. Accurate interval-energy prediction can support household energy analysis and the design of demand-management tools, although reductions in bills or energy use require separate deployment evidence.

Edwards et al. [1] examined next-hour residential consumption, while Candanedo et al. [2] modelled appliance energy using indoor environmental measurements and weather observations. Reviews of building energy prediction [3, 4] show that model performance depends on the target, input information, observation scale, and evaluation design. These studies motivate task-specific assessment rather than a universal ranking of learning algorithms.

Tree ensembles can represent nonlinear relationships and interactions in tabular telemetry. Combining randomized trees with boosting methods may be useful when their prediction errors contain complementary information. Stacking must nevertheless demonstrate an improvement over tuned individual learners on the same held-out observations. An increase in architectural complexity alone does not establish a predictive contribution.

The present task is energy prediction for an observation interval. Contemporaneous regression estimates that interval using measurements from the same interval; forecasting predicts a future interval using information available beforehand. A ten-minute sampling frequency does not itself define a ten-minute forecast horizon. Prospective forecasting is therefore reserved for a subsequent evaluation with explicit predictor availability and lead time.

Anomaly screening provides a separate analytical output. An observation may be unusual in sensor feature space, associated with a large prediction error, or both. None of these conditions independently establish

appliance failure: an ordinary but infrequent activity, a sensor error, or model misspecification can also generate an alert. Independent event labels are necessary for physical diagnosis and detection-performance estimates.

(500 words with 7-8 papers)

2. Literature survey

2.1. Residential energy modelling

Early residential studies demonstrate the importance of matching the modelling task to the observation scale. Edwards et al. [1] investigated next-hour consumption, and Candanedo et al. [2] developed the appliance-energy dataset used in this study. Their unrestricted target setting differs from the nominal-demand subset examined here. Fan et al. [5] combined outlier analysis, feature selection, and ensembles for next-day building energy and peak-demand prediction. Ensemble construction and preprocessing are therefore established approaches, and their use alone does not establish novelty.

Ahmad and Chen [6] examined heating and cooling loads, while Olu-Ajayi et al. [7] compared deep learning with other techniques for residential building energy prediction. These studies inform model design, but their targets and observation scales differ from ten-minute aggregate appliance energy. Chou and Hsu [8] integrated web-based acquisition with optimized prediction, illustrating the distinction between software integration and validation of predictive performance.

Temporal architecture offers another modelling route. Saad Saoud et al. [9] combined the stationary wavelet transformation with Transformers. Wang et al. [10] integrated cumulative temperature effects, decomposition, recurrent modelling, and XGBoost residual correction. Pai H et al. [11] examined weather-enriched multivariate long short-term memory models, whose recurrent-memory foundation was introduced by Hochreiter and Schmidhuber [12]. A valid comparison with tree ensembles requires harmonized targets, predictor availability, preprocessing, and held-out timestamps.

Several studies emphasize household context. Yang et al. [13] considered consistent modelling across households with privacy preservation. Qavideh-Fard et al. [14] incorporated occupant-related parameters, which provide information beyond calendar proxies. Ramnath et al. [15] combined questionnaire and monthly-consumption data for 225 consumers in Maharashtra. Neo et al. [16] included building characteristics and urban landscape information in Singapore. Differences in population, geography, resolution, and predictors prevent these results from serving as direct numerical benchmarks for the present experiment.

2.2. Explainability and application integration

Cui et al. [17] examined household features using machine learning and SHAP. Feature attribution can clarify model behaviour, but an importance ranking does not establish a causal effect on consumption. Al-Rajab and Loucif [18] developed Sustainable EnergySense with appliance identification and user-facing energy information. Such application designs motivate interface integration without independently establishing energy savings or the validity of anomaly labels.

Abd'Azeez and Olatomiwa [19] reported an ExtraTrees appliance-energy model with an API, including $R^2 = 0.7441$ and $RMSE = 11.75$ Wh. This study is relevant context. Matching a dataset name, however, does not establish identical filtering, partitioning, predictor availability, or tuning; the published values are not used to rank the present stack.

600 words

Table 1 summarizes representative studies and the conditions that limit direct comparison with the current ten-minute prediction task.

Table 1
Selected studies and their relationship to the present task

Study	Main emphasis	Comparison constraint
Candanedo et al. [2]	Indoor and weather inputs for appliance energy	Unrestricted target scope differs from nominal-range evaluation
Fan et al. [5]	Data preparation and ensemble modelling	Next-day building energy and peak demand
Saad Saoud et al. [9]	Wavelet features and Transformers	Temporal preprocessing and horizon must be harmonized
Ramnath et al. [15]	Survey and monthly-consumption data	Different population and temporal resolution
Cui et al. [17]	Prediction and SHAP analysis	Attributions describe models rather than causal effects
Abd'Azeez and Olatomiwa [19]	ExtraTrees prediction with an API	Requires reproduction on identical test observations

2.3. Research gap (3-4 points)

The evaluation questions are whether stacking improves on tuned base learners on identical observations; whether cyclic and environmental descriptors add information beyond raw inputs and temporal baselines; whether screening produces useful alerts after false alarms are assessed; and whether the complete pipeline generalizes across time, target ranges, households, and deployment conditions. These are testable objectives. The preliminary experiment does not yet answer them conclusively.

Study contributions (3-4 points)

The study examines whether heterogeneous tree ensembles, environmental descriptors, and residual monitoring can form a reproducible workflow for household energy analysis. Its contributions are integrated prediction architecture, a separation of feature-space novelty from residual exceedance, and a preliminary interpretation of prediction and screening diagnostics. The present evidence concerns one dwelling and a restricted target range; a controlled comparison remains necessary to quantify the contribution of stacking.

The paper is organized as follows. Section 2 reviews related research. Section 3 describes the regressors and stacking formulation. Section 4 presents the dataset, feature representation, model settings, evaluation metrics, and screening protocol. Section 5 discusses preliminary results, diagnostic figures, prototype integration, and limitations. Section 6 concludes the study and identifies the required validation work.

3. Machine learning regressors and ensemble formulation

3.1. Extremely randomized trees

Extremely randomized trees, or ExtraTrees (ET), introduce random candidate split thresholds and aggregate multiple trees [20]. For M fitted trees with predictions $f_m(x)$, regression uses their arithmetic mean:

$$\hat{y}_{\text{ET}}(x) = \frac{1}{M} \sum_{m=1}^M f_m(x) \quad (1)$$

The documented configuration uses 400 trees. Randomization can reduce similarity among trees, but its effect on generalization depends on the feature representation and fitting settings. Random forests [21] provide a related baseline for a future controlled comparison.

3.2. Extreme gradient boosting

Gradient boosting builds an additive predictor through successive corrections to a loss function [22]. XGBoost adds regularization and an efficient tree-learning implementation [23]. Its additive prediction can be expressed as:

$$\hat{y}_{\text{XGB}}(x) = f_0(x) + \eta \sum_{m=1}^M f_m(x) \quad (2)$$

Here f_0 is the initial prediction and η denotes shrinkage. The reported configuration specifies 700 estimators, learning rate 0.03, maximum depth 8, row and column subsampling of 0.85, and regularization parameters of 0.5 and 1.0. These are documented settings rather than evidence of a completed hyperparameter search.

3.3. Histogram gradient boosting

Histogram gradient boosting (HGB) assigns continuous predictor values to bins before searching for tree splits. This approximates split evaluation while retaining the additive boosting structure. The documented configuration specifies 500 maximum iterations, 255 maximum bins, learning rate 0.04, and maximum depth 9. Early stopping behavior, validation handling, and other estimator settings remain unspecified. Versioning the software and preprocessing together is necessary to reproduce the fitted pipeline [24].

3.4. Non-negative ridge stacking

Stacked generalization combines base learners using held-out predictions rather than predictions fitted to the same training targets [25, 26]. Let Z contain the ET, XGB, and HGB prediction columns and let y contain the corresponding measured energies. The documented formulation applies ridge penalization [27] with non-negative coefficients:

$$w^* = \arg \min_{w \geq 0} \{ \|y - Zw\|_2^2 + \alpha \|w\|_2^2 \}, \quad \alpha = 1.0 \quad (3)$$

$$\hat{y} = w_{\text{ET}} \hat{y}_{\text{ET}} + w_{\text{XGB}} \hat{y}_{\text{XGB}} + w_{\text{HGB}} \hat{y}_{\text{HGB}} \quad (4)$$

Equations (3) and (4) describe the no-intercept formulation. An intercept must be added to both the fitted-model description and prediction equation if it is present in the saved implementation. Non-negativity prevents subtractive combinations but does not impose a sum-to-one constraint. The reported values 0.481, 0.343, and 0.176 sum to one; whether they are actual fitted coefficients or normalized display weights remain unresolved. Prediction must use the fitted coefficients rather than inferred normalized weights.

Write equations in word and make attractive flowchart of the whole study

Fig. 1 illustrates the documented architecture. The feature dimension is left unspecified because the complete list corresponding to the stated 50 inputs has not been established.

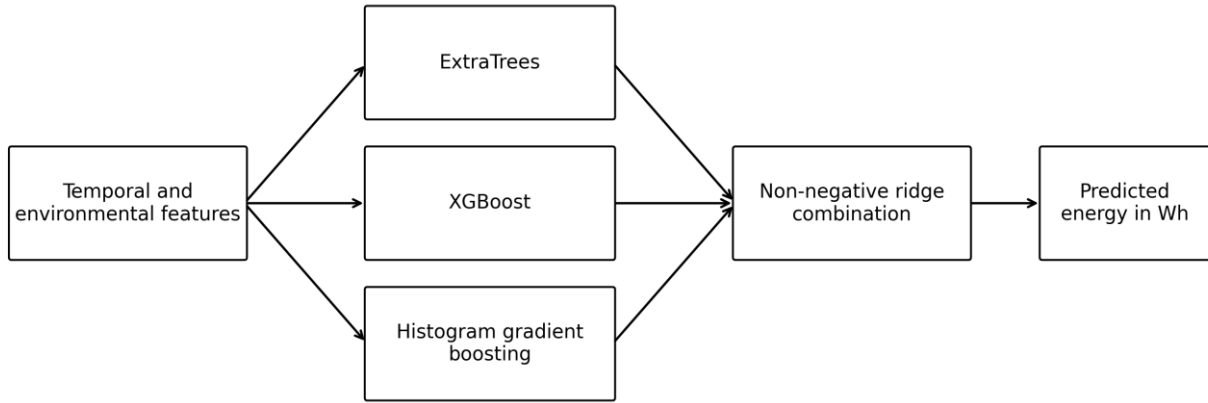


Fig. 1. Prediction architecture with three base regressors and a non-negative ridge combination. The exact input schema and fitted coefficient conventions require confirmation.

4. Materials and methodology

4.1. Dataset description

The Appliances Energy Prediction dataset is available from the UCI Machine Learning Repository [28] and is documented by Candanedo et al. [2]. It contains ten-minute observations from one low-energy dwelling in Stambruges, Belgium. The source-data audit reports 19,735 records and 29 CSV columns, covering 11 January 2016 at 17:00 to 27 May 2016 at 18:00, with no missing fields or duplicate timestamps. The 29-column total includes the target; it should not be interpreted as 29 predictor features.

Appliances are aggregate appliance energy in watt-hours per ten-minute interval. It is neither instantaneous power nor an appliance-specific measurement. The documented target range is 10-1,080 Wh, with mean approximately 97.69 Wh and median 60 Wh. Lighting energy is stored separately. The recorded timezone is not established, so the observations are not labelled UTC. **Source link**

Table 2 groups the source variables by measurement role. Sensor 6 is outdoors and is excluded from indoor-only aggregates.

Table 2
Dataset variables and measurement interpretation

Group	Variables	Interpretation
Target	Appliances	Aggregate appliance energy in Wh per ten-minute interval
Lighting	lights	Separately recorded lighting energy in Wh
Indoor sensors	T1-T5, T7-T9; RH_1-RH_5, RH_7-RH_9	Room temperature in °C and relative humidity in %
Outdoor wall sensors	T6; RH_6	North-side outdoor temperature and humidity
Weather station	T_out, Press_mm_hg, RH_out, Windspeed, Visibility, Tdewpoint	Outdoor temperature, pressure, humidity, wind, visibility, and dew point
Other columns	date; rv1; rv2	Timestamp and two random control variables

4.2. Target filtering and evaluation population

The documented nominal-demand experiment removes rv1 and rv2 and retains observations satisfying $10 \leq \text{Appliances} \leq 120$ Wh. This target-range restriction is not an interquartile-range filter. It retains 16,740 observations, or 84.82% of the source series, and excludes 2,995 observations. Performance on this subset concerns conditional low-to-moderate demand; it does not establish unrestricted or peak-load performance.

Table 3 records the population and split counts. The counts determine the stated 70/30 proportions but do not establish whether the partition was random or chronological.

Table 3
Observation counts and documented evaluation scope

Population	Count	Interpretation
Complete source series	19,735	One dwelling at ten-minute resolution
Nominal target range	16,740	10-120 Wh; 84.82% of source observations
Excluded observations	2,995	15.18% of source observations
Stated training sample	11,718	70% of retained observations
Stated test sample	5,022	30% of retained observations; partition unresolved

Under a chronological split after filtering, the training portion ends on 17 April 2016 at 18:30 and testing begins at 18:40. The reported metrics are not consistent with that exact test-target distribution, as shown in Section 5.1. A claim of chronological generalization is therefore not supported by the present record.

A prospective evaluation should divide the original series into training, validation, and test periods before fitting any data-dependent transformation. Target filtering also uses measured energy, which is unknown when prospective predictions are requested. Full-range test evaluation is therefore necessary, with any nominal-range analysis reported as a separate conditional assessment. Removing observations breaks the regular interval sequence; lag construction must preserve actual timestamp differences.

4.3. Temporal and environmental features

Cyclic encoding represents adjacency across periodic boundaries. For an index u with period P , sine and cosine descriptors are defined as:

$$c_1(u) = \sin(2\pi u/P), \quad c_2(u) = \cos(2\pi u/P) \quad (5)$$

The documented periods are 24 for hour, 144 for ten-minute slot, 7 for weekday, and 12 for month. The January-May record cannot validate annual seasonality. Morning, evening, night, and weekend indicators are calendar proxies rather than observed occupancy. The stated hour windows are 07:00-10:00, 18:00-22:00, and 01:00-05:00; inclusivity of each endpoint must be recorded in the implementation.

Let $I = \{1, 2, 3, 4, 5, 7, 8, 9\}$ denote the eight indoor sensor indices. Indoor means, temperature spread, and environmental differences are defined as:

$$\bar{T}_{\text{in}} = \frac{1}{8} \sum_{i \in I} T_i, \quad \overline{RH}_{\text{in}} = \frac{1}{8} \sum_{i \in I} RH_i \quad (6)$$

$$T_{\text{spread}} = \max_{i \in I} T_i - \min_{i \in I} T_i \quad (7)$$

$$\Delta T_{\text{out}} = \bar{T}_{\text{in}} - T_{\text{out}}, \quad D_{\text{dew}} = T_{\text{out}} - T_{\text{dewpoint}} \quad (8)$$

$$\Delta T_{\text{living}} = T_2 - T_{\text{out}} \quad (9)$$

These quantities describe environmental conditions; they do not constitute a physical heat-transfer model. Averaging all nine humidity channels would produce a mixed indoor-outdoor quantity. The stated input count of 50 is not fully explained by the documented transformations. Reproduction requires the retained raw columns, additional aggregates, exact feature order, treatment of lighting energy, and all preprocessing parameters. Lighting information must also be available at the intended prediction time before it can be accepted as prospective input.

4.4. Documented model configuration

Table 4 summarizes the documented configuration. The reported settings must be distinguished from tuned settings because search spaces, selection criteria, and completed tuning results are not established.

Table 4
Model settings and remaining implementation details

Component	Documented settings	Unresolved details
ET	400 trees	Other parameters and random seed
XGB	700 estimators; learning rate 0.03; depth 8; subsample 0.85; colsample_bytree 0.85; reg_alpha 0.5; reg_lambda 1.0	Objective, seed, software version, and remaining settings
HGB	500 maximum iterations; 255 maximum bins; learning rate 0.04; depth 9	Early stopping, validation handling, and remaining settings
Ridge layer	Non-negative coefficients; $\alpha = 1.0$	Intercept and coefficient normalization
Isolation Forest	250 trees; contamination 0.025	Fitting population, seed, and remaining settings

4.5. Temporal validation and stacking protocol

Time-series validation depends on the prediction task and dependence structure [29]. A reproducible temporal stack should generate Z through expanding or rolling training windows within the training period. Each validation block must follow the fitting block that generates its predictions. The warm-up block, which has no earlier out-of-fold prediction, is excluded from meta-model fitting. After selection, base learners can be refitted on the complete training period and the stack evaluated once on untouched test observations.

The current record does not establish the completed fold assignments. Hyperparameter search, model selection, imputation, any fitted scaling, and detector calibration must remain confined to training or earlier validation data. Simple mean, persistence, and seasonal-naïve baselines should be selected according to input availability. Their predictions and those of tuned ET, XGB, HGB, and the stack must be assessed at the same test timestamps.

Fig. 2 summarizes the workflow required to reproduce and validate the architecture. The test period must not determine fitted transformations, stacking coefficients, or anomaly thresholds.

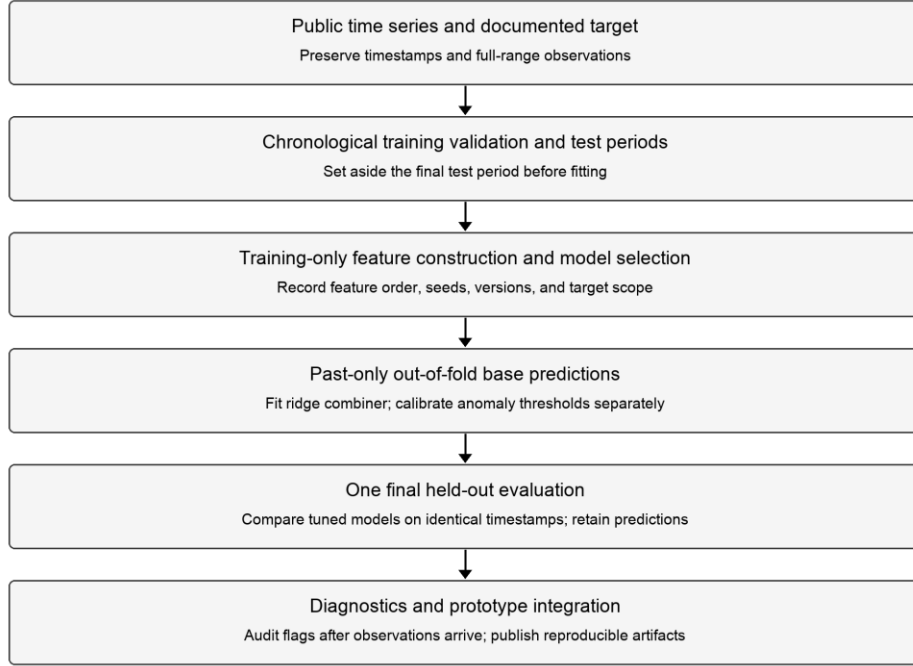


Fig. 2. Proposed verification workflow separating training, model selection, calibration, and final testing. It is a validation design rather than evidence that the preliminary experiment completed each step. (short it)

4.6. Performance evaluation metrics

For n test observations, measured interval energy y_i , prediction $\hat{y}_i$, and test-target mean $\bar{y}$, mean absolute error (MAE), root mean squared error (RMSE), the coefficient of determination (R^2), and mean absolute percentage error (MAPE) are defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (10)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (12)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (13)$$

MAE and RMSE are expressed in Wh, R^2 is dimensionless, and MAPE is expressed as a percentage. R^2 is not classification accuracy. MAPE is defined for the retained positive targets but gives relatively greater weight to low-demand errors [30]. All compared metrics must use identical test observations, weighting conventions, and target transformations.

4.7. Anomaly screening and threshold calibration

Isolation Forest identifies unusual feature vectors through randomized partitioning [31]. Its contamination parameter of 0.025 controls the fitted score threshold; it does not establish a 2.5% fault prevalence or prescribe

a combined flag rate. Feature-space novelty and residual exceedance are distinct signals and should be reported separately.

Residual screening requires measured energy to become available. Let r_i denote the signed residual and e_i its absolute magnitude:

$$r_i = y_i - \hat{y}_i, \quad e_i = |r_i| \quad (14)$$

One documented rule applies an upper fence to calibration magnitudes e_{cal} , where Q_1 and Q_3 are the first and third quartiles:

$$\tau = Q_3(e_{\text{cal}}) + 3[Q_3(e_{\text{cal}}) - Q_1(e_{\text{cal}})], \quad e_i > \tau \quad (15)$$

A separate signed-residual rule uses lower and upper fences:

$$\tau_L = Q_1(r_{\text{cal}}) - 3 \text{IQR}(r_{\text{cal}}), \quad \tau_U = Q_3(r_{\text{cal}}) + 3 \text{IQR}(r_{\text{cal}}) \quad (16)$$

Calibration residuals should come from an earlier validation period or time-respecting out-of-fold predictions. Test residuals must not set the threshold used to assess that same test period. The signed cutoffs shown in Fig. 4 have not been established as equivalent to the magnitude rule in Equation (15). A fixed cutoff is not adaptive unless a documented past-only update rule changes over time.

The combination of Isolation Forest and residual screening must be specified as a union, intersection, or staged rule. A prediction endpoint cannot calculate a residual flag from environmental inputs alone. High positive residual or night-time alert are descriptive labels; standby waste, runaway consumption, and appliance faults require independent evidence.

5. Experimental results and discussion

5.1. Preliminary nominal-demand performance

Table 5 presents the reported stacked-model metrics for the nominal-demand experiment. These values remain preliminary because the evaluation partition has not been reconciled.

Table 5
Reported preliminary metrics for the stacked ensemble

Evaluation scope	R ²	MAE Wh	RMSE Wh	MAPE %
10-120 Wh; stated n = 5,022; partition unresolved	0.7504	8.62	11.58	15.53

For the unweighted definitions in Equations (11) and (12), the population standard deviation of the test targets is determined by the RMSE and R² pair:

$$\sigma_y = \frac{\text{RMSE}}{\sqrt{1-R^2}} \quad (17)$$

The reported values imply $\sigma_y \approx 23.18$ Wh. The source-data audit gives approximately 21.64 Wh for the chronological final 5,022 observations of the filtered series. This difference exceeds the effect of the reported rounding. The partition, target transformation, weighting convention, or numerical results must therefore be reconciled before the metrics constitute a reproducible evaluation.

The initially reported baseline metric pairs imply target standard deviations near 100.03-100.04 Wh, which differ substantially from the stack's implied value. These results may describe different target populations and are excluded from a common-test ranking. A changed target range cannot be attributed to an improvement in model architecture.

Fig. 3 shows prediction dispersion around the identity line. Some low-demand observations are overpredicted and some higher nominal-demand observations are underpredicted.

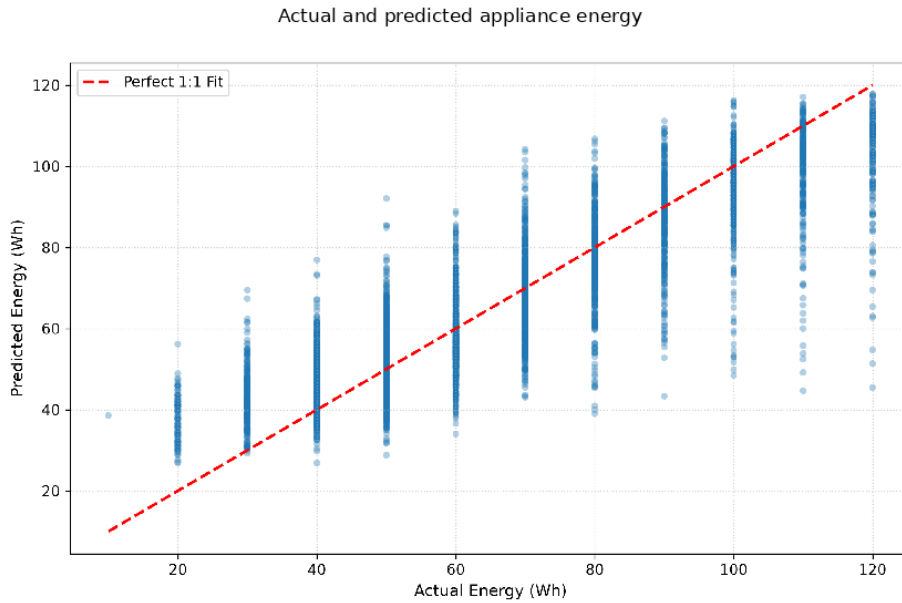


Fig. 3. Actual-versus-predicted energy in the nominal range. The plotted points are retained from the preliminary experiment; evaluation consistency is discussed in Section 5.1. **(short)**

The scatter supports examining conditional errors, but does not establish uniform accuracy or performance on excluded high-demand intervals. A future evaluation should report error by demand band and time of day, together with timestamped predictions.

5.2. Residual behaviour

Fig. 4 summarizes the signed residuals. Most displayed errors are near zero, with tails extending in both directions.

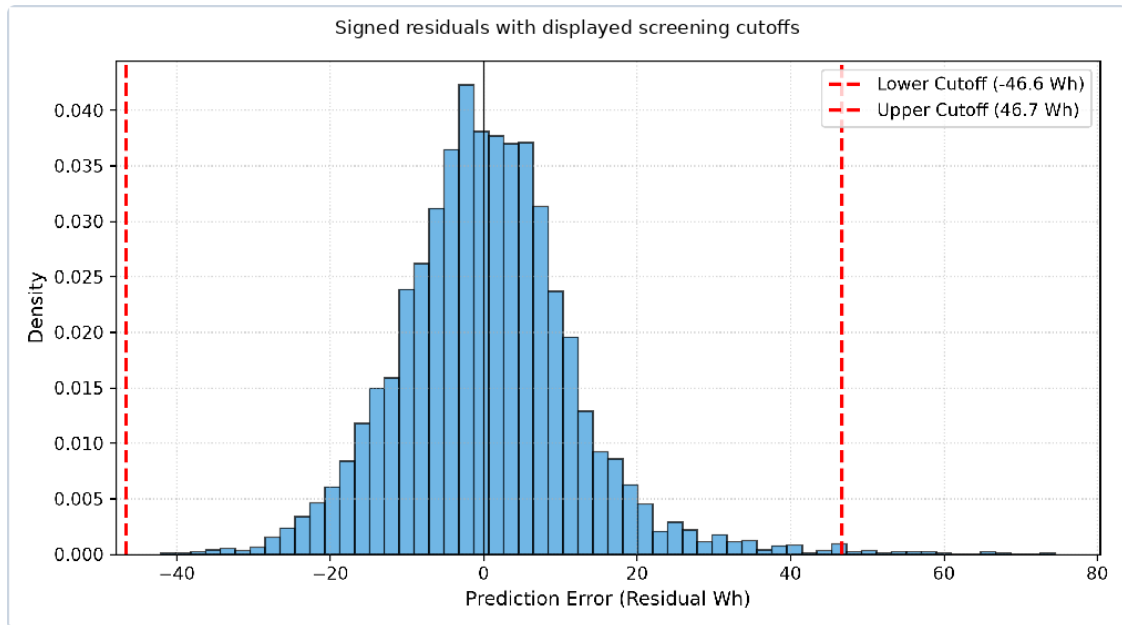


Fig. 4. Signed-residual histogram with displayed cutoffs of approximately -46.6 and +46.7 Wh. The calibration population and relationship to Equations (15) and (16) require confirmation.

The histogram alone does not establish normality, conditional unbiasedness, or temporal independence. Residuals should also be examined against time, fitted energy, recorded hour, and demand range. The displayed boundaries are screening cutoffs; an adaptive update mechanism has not been established.

5.3. Feature importance and interpretability

Fig. 5 ranks 15 displayed features. Time_slot_sin, Hour_sin, is_evening_peak, Hour, and Time_slot lead the ranking, followed by T8.

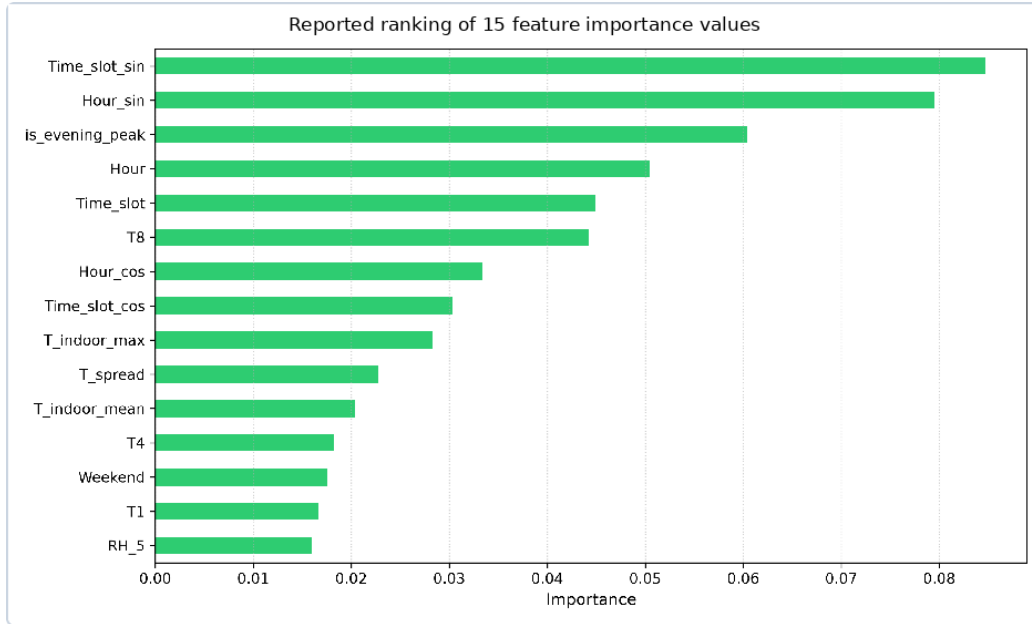


Fig. 5. Reported feature-importance ranking. The estimator, importance definition, and aggregation method have not been specified.

The chart does not list T2 among its displayed features. The ranking cannot be assigned to the complete stack without the generating method, and a regression importance calculation should not be called Gini importance. Correlated temporal descriptors can distribute importance among related variables. SHAP can supplement an explicitly documented importance analysis [32], while remaining an explanation of model predictions rather than evidence of causal effects.

5.4. Flagged observations and temporal distribution

Fig. 6 highlights 24 flagged observations. With the stated denominator of 5,022, the flag rate is $24/5,022 \times 100 = 0.478\%$, rounded to 0.48%.

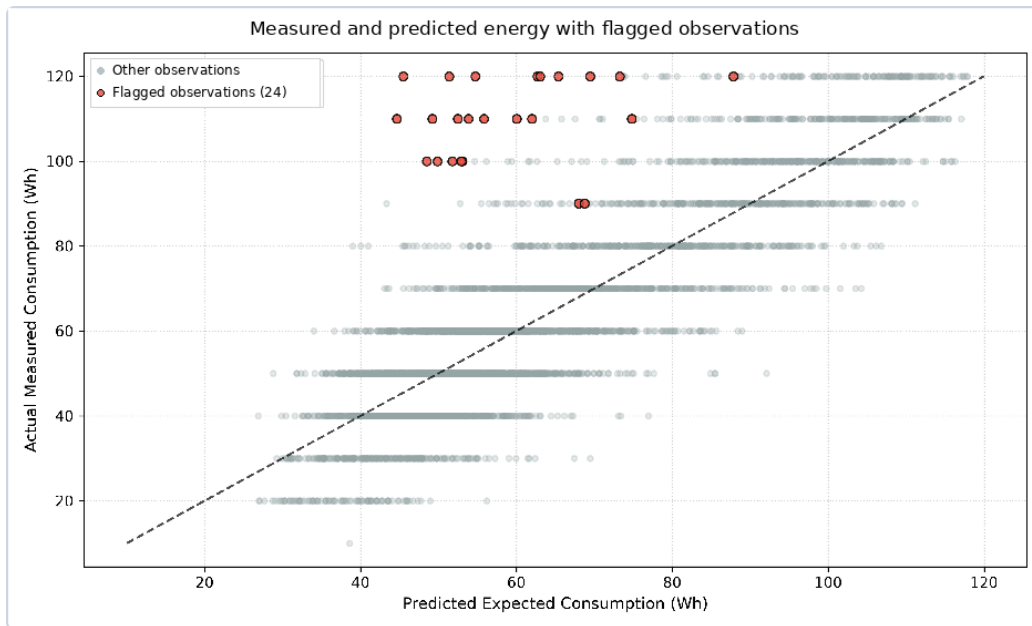


Fig. 6. Measured and predicted nominal-demand energy with 24 flagged observations. Flags are screening outputs rather than verified appliance faults.

This proportion is not detection accuracy or fault prevalence. The measured plotted targets remain within the nominal range and do not establish consumption above 250 Wh. Classifying 23 observations as runaway events and one as standby waste is not supported by independent event labels.

Fig. 7 shows the hourly distribution of flags. Five occur at hour 6, five at hour 7, and four at hour 8; these hours account for 14 of the 24 flags.

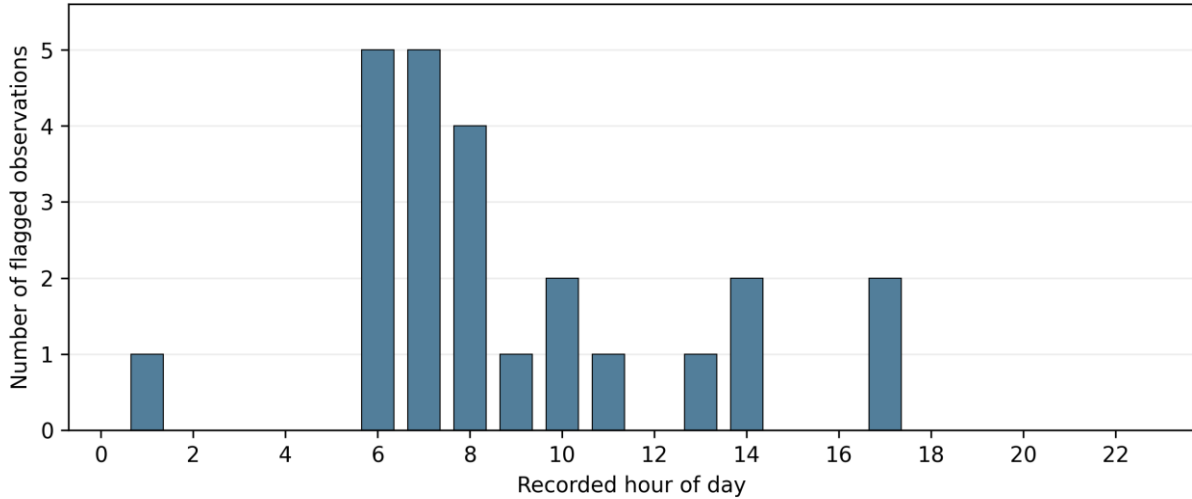


Fig. 7. Counts of flagged observations by recorded hour. Bars reproduce the aggregate counts in the preliminary diagnostic figure; the total is 24.

The remaining counts are one each at hours 1, 9, 11, and 13, and two each at hours 10, 14, and 17. The displayed evidence does not establish an evening concentration or a specific 03:20 event. Timestamped records are required to identify consecutive alerts and determine whether multiple flagged intervals belong to one episode.

5.5. Comparative evaluation and uncertainty

The ExtraTrees results reported by Abd’Azeez and Olatomiwa [19] are published context rather than a controlled baseline. Differences between their reported metrics and those in Table 5 are descriptive. Neither statistical significance nor state-of-the-art performance is established. The baseline must be reproduced on the same inputs and test timestamps before any benefit can be attributed to stacking.

A previous cross-dataset comparison changes household population, geography, predictors, units, temporal resolution, and sample size simultaneously. It cannot isolate a causal effect of data volume. The associated survey scores and ANOVA significance claim are therefore not used to support the present conclusions. A controlled learning curve must vary training-set size while preserving features and held-out periods.

Future ablations should compare raw variables, calendar terms, cyclic encodings, and environmental aggregates. Repeated temporal origins or dependence-aware resampling are needed to quantify uncertainty in performance differences. Adjacent ten-minute observations should not be treated as independent experimental replicates. A negative test R^2 indicates worse squared error than the test-target mean reference; it does not independently identify an overfitting mechanism.

5.6. Prototype integration

The documented prototype design serializes the prediction pipeline and screening model and exposes them through a Flask REST service. The proposed endpoints are `/api/predict`, `/api/anomaly_audit`, and `/api/cross_experiments`. Endpoint latency, reliability, and production operation have not been established by the reported evaluation.

Fig. 8 illustrates the intended interface connection. Prediction responses should include interval energy, model version, and input-validation status; residual auditing additionally requires measured energy and its timestamp.



Prediction → JSON response → frontend visualization

Fig. 8. Proposed frontend, REST-service, and serialized-model connection. Prediction and retrospective residual auditing require different request inputs.

Inference must preserve the fitted feature definitions, order, measurement units, and transformations. A small number of dashboard sliders cannot replace missing sensor channels without a validated imputation or scenario-construction rule. Responses should distinguish feature-space and residual flags. The proposed fallback threshold of 1,000 records is uncalibrated and cannot be presented as a general rule. Model size, memory use, malformed-input handling, endpoint consistency, and response latency remain practical evaluation targets.

5.7. Study limitations and reproducibility

The study covers one dwelling over approximately four and a half months. It does not establish transfer across households, climates, appliance inventories, or a complete annual cycle. The target filter excludes 15.18% of source observations, including important high-demand behaviour. Contemporaneous measurements do not establish a prospective forecasting capability.

Table 6 identifies the records needed to turn the preliminary assessment into a reproducible evaluation. These requirements concern the underlying experiment and cannot be resolved through editorial changes alone.

Table 6
Evidence required to resolve the principal reproducibility limitations

Element	Current limitation	Required evidence
Evaluation partition	Counts without a reconciled split	Timestamped train/test assignments and regenerated predictions
Feature representation	Stated 50 inputs are incompletely defined	Ordered schema, units, transformations, and lighting treatment
Stacking	Fold construction and coefficient handling unresolved	Out-of-fold indices, fitted coefficients, and intercept convention
Anomaly screening	Calibration and combined decision rule unresolved	Independent calibration records and explicit signal combination
Comparison	No matched baseline or uncertainty analysis	Tuned learners on identical test timestamps and temporal uncertainty
Fault interpretation	No independent event labels	Verified events and event-level precision and recall

No controlled ablation, confidence interval, or matched baseline establishes the improvement attributable to stacking. Feature interpretation is constrained by an unspecified importance calculation and correlated inputs. Independent labels are absent for anomaly precision, recall, F1 score, and fault diagnosis. Energy savings, user benefit, and operational reliability require separate validation after the methodological issues are resolved.

6. Conclusion and future work

This study examines an integrated architecture for nominal-range residential appliance energy prediction and retrospective anomaly screening. ET, XGB, and HGB regressors use temporal and environmental descriptors, followed by a non-negative ridge combination. The source-data audit reports 19,735 ten-minute observations, including 16,740 records within the 10-120 Wh interval. Preliminary results are $R^2 = 0.7504$, $MAE = 8.62$ Wh, $RMSE = 11.58$ Wh, and $MAPE = 15.53\%$, subject to reconciliation of the evaluation partition and metric consistency.

The diagnostic figures describe prediction dispersion, residual tails, a feature ranking, and 24 flagged observations. They support analysis of model behaviour without establishing appliance faults or standby-energy waste. The hourly counts concentrate 14 of the 24 flags at recorded hours 6-8. A proposed REST architecture provides an integration route for prediction and auditing, with separate input requirements for each operation.

The present contribution is an integrated system design with a preliminary assessment and explicit evaluation boundaries. Future work should publish the feature schema and split assignments, generate time-respecting out-of-fold predictions, compare tuned learners on identical timestamps, and calibrate screening independently. Full-range and multi-household evaluation, uncertainty estimation, verified event labels, and reproducible software artifacts are needed before claims of operational forecasting, comparative superiority, or household energy savings can be supported.

Data availability

The source dataset is publicly available from the UCI Machine Learning Repository [28] at <https://doi.org/10.24432/C5VC8G> under the Creative Commons Attribution 4.0 licence. A project-specific reproducibility package containing the fitted model, ordered feature schema, split assignments, calibration records, and timestamped predictions has not been made available with the manuscript.

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5. Experimental Results and Empirical Evaluation

5.1. Preliminary Predictive Performance Under Nominal-Demand Conditions

This is better than “Preliminary nominal-demand performance” because it identifies **what was evaluated** and **under which operating condition**.

5.2. Residual Structure and Prediction-Error Characteristics

This is stronger than simply “Residual behaviour” and allows you to discuss distribution, tails, temporal dependence, heteroscedasticity, and bias.

5.3. Feature Importance and Model Interpretability

rather than claiming SHAP when the current importance-generation procedure is unresolved.

5.4. Anomaly Screening and Temporal Distribution of Flagged Observations

This is more precise than “Flagged observations and temporal distribution” and avoids implying that the flags are verified faults.

5.5. Comparative Benchmarking and Uncertainty Assessment

This appropriately captures:

- comparison with previous papers
- matched versus cross-dataset comparisons;
- learning curves;
- ablation studies;
- temporal uncertainty;
- statistical dependence.

5.6. Prototype Deployment Architecture and Operational Readiness

5.7. Reproducibility, Evidence Gaps, and Study Limitations

6. Discussion

6.1. Reliability of the Reported Predictive Performance

Discuss the unresolved evaluation partition, target-population inconsistency, and why the current R^2 /MAE/RMSE values cannot yet support a definitive performance claim.

6.2. Implications of Evaluation-Partition and Target-Distribution Inconsistencies

This can become an important methodological contribution. Explain why apparently different target distributions can change R^2 and why numerical reconciliation is necessary before comparing models.

6.3. Interpretability of Temporal and Environmental Predictors

Discuss the dominance of temporal variables while explicitly avoiding causal interpretation.

6.4. Anomaly Screening Versus Verified Fault Detection

The 24 flags should be presented as **screening candidates**, not confirmed runaway events or appliance faults.

6.5. Requirements for Reliable Energy-Forecasting and Anomaly-Detection Systems

Connect the findings to:

- timestamped evaluation;
- prospective forecasting;
- independent fault labels;
- uncertainty estimation;
- matched baselines;
- temporal validation.

6.6. Generalizability and External Validity

Discuss the single dwelling, 4.5-month observation period, excluded high-demand observations, household variability, climate, appliance inventories, and annual seasonality.

6.7. Reproducibility and Deployment Implications

Discuss the need for:

- explicit feature schema;
- reproducible stacking folds;
- calibrated anomaly thresholds;
- model/version tracking;
- API validation;

- latency/reliability testing.